

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

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Current Agent Training Uses "Flood Irrigation"

Trajectory Imitation

ReAct and similar methods directly copy gold trajectories without situational filtering.

Example: ReAct

Trial-and-Error

Reflexion injects external feedback regardless of whether the agent actually needs it.

Example: Reflexion

Knowledge-Augmented

KnowAgent, WKM flood domain knowledge at every step — 100% knowledge injection.

Example: KnowAgent, WKM

The "Flood Irrigation" Problem

All three paradigms indiscriminately inject resources regardless of situational need → fragile planning, pattern collapse, high inference costs.

Human Metacognition: The Missing Piece

Humans possess **situational self-awareness**— the ability to assess at each decision point:

Act Directly

When the situation is familiar and the path is clear.

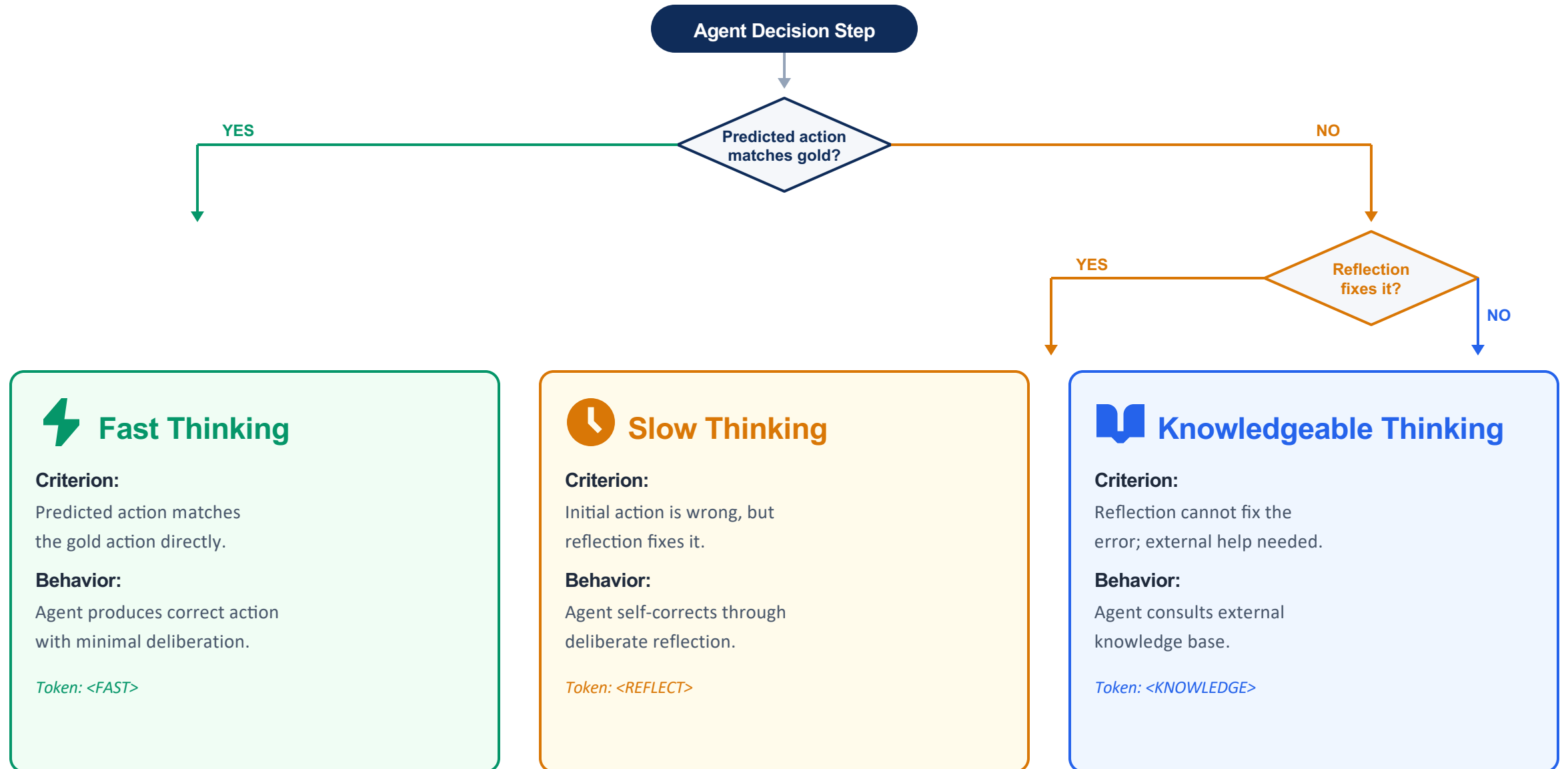
Pause & Reflect

When the first instinct is wrong but self-correction is possible.

Seek External Help

When reflection fails and external knowledge is required.

Three Situational Modes



Step 1 — Data Construction

Self-Awareness Labels

- Agent self-explores environments
- Heuristic criterion classifies each step:
 - Match gold → <FAST>
 - Reflection fixes → <REFLECT>
 - Otherwise → <KNOWLEDGE>

No human annotation required

Step 2 — Two-Stage Training

Stage A: SFT

Supervised fine-tuning teaches initial self-awareness patterns from labeled data.

Stage B: RPO

Reward-based Policy Optimization refines the agent's special-token generation.

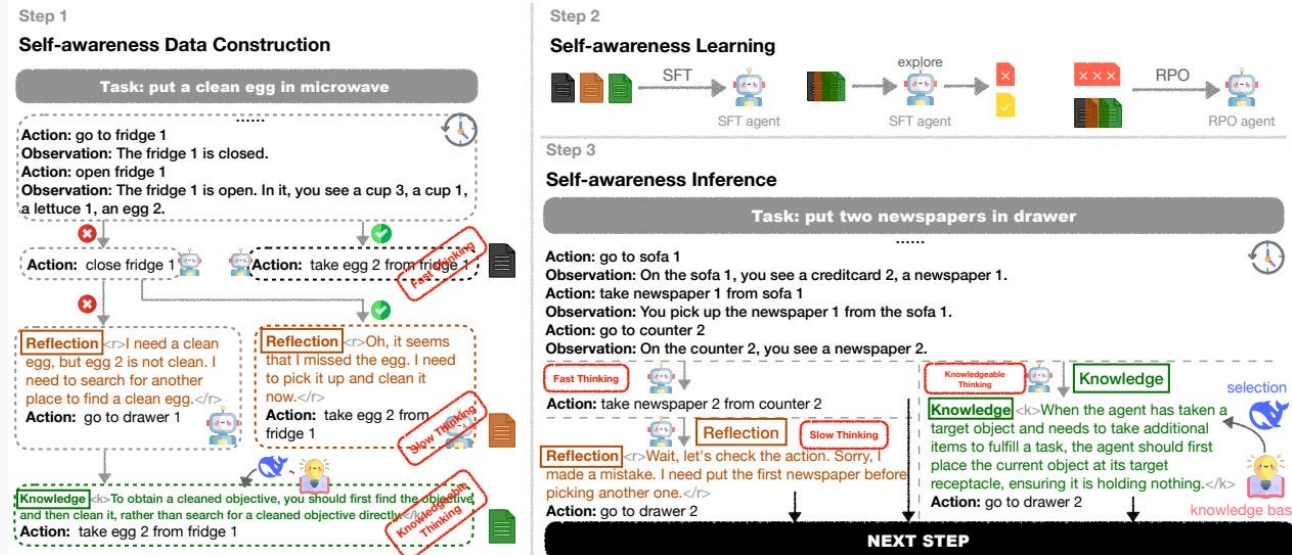
Step 3 — Self-Aware Inference

Token-Gated Execution

- <FAST> → direct action output
- <REFLECT> → reflection loop then act
- <KNOWLEDGE> → retrieve from KB then act

Deployment-time situational self-awareness

KnowSelf Pipeline Overview



Qualitative Examples: KnowSelf vs. Baselines

Example A: "Put a hot mug in cabinet"

OpenAI-O1 (Baseline)

Opens the microwave, then incorrectly tries to take an **egg** from inside — failing the task.

No situational awareness of what needs heating.



KnowSelf (Llama-8B)

Emits a **<REFLECT>** token, realizes it should **heat the mug** using the open microwave.

Self-awareness triggers correct reflection.



Insight:

KnowSelf recognizes the need for reflection and self-corrects, while the baseline hallucinates an irrelevant object.

Example B: "Put a saltshaker in drawer"

DeepSeek-R1 (Baseline)

Reflects that the drawer is open and tries to **close it first** — which fails the task logic.

Reflection without proper knowledge leads to error.



KnowSelf

Applies a **<KNOWLEDGE>** token noting the agent should **place the object** without removing unrelated items.

Knowledge retrieval resolves the ambiguity.



Insight:

KnowSelf knows when reflection is insufficient and correctly invokes external knowledge instead of guessing.

ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge

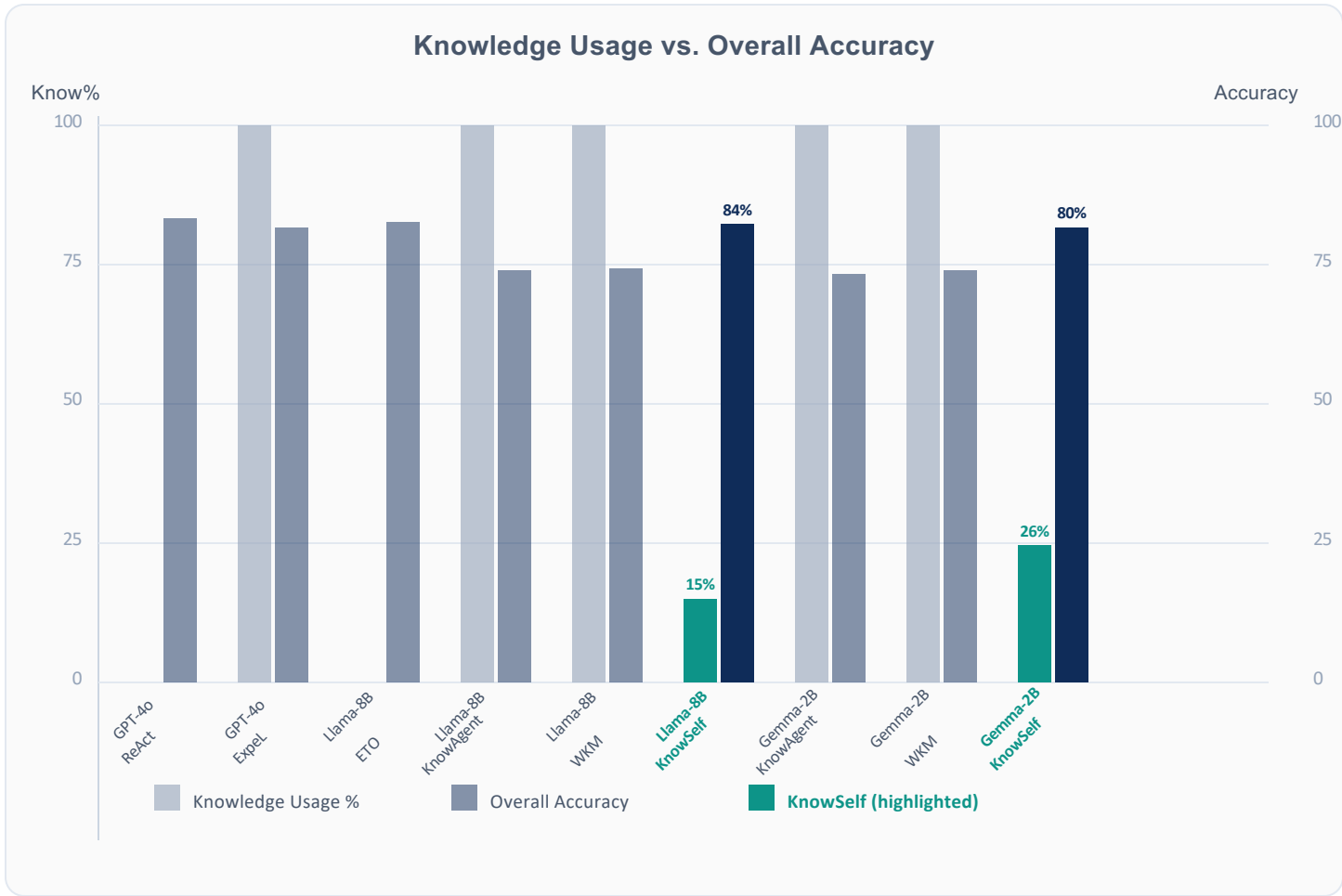
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85



Key Insight

- KnowSelf (Llama-8B) achieves**84.33%**overall — surpassing GPT-4o ExpeL (**79.85%**) while using only**15%**knowledge.
- Gemma-2B KnowSelf reaches**79.85%**, matching GPT-4o ExpeL — selective knowledge scales down effectively.

Minimal Knowledge, Maximum Impact



Efficiency Insights

~6× Knowledge Reduction

Llama-8B KnowSelf uses only **15%** knowledge vs. 100% for Expel, KnowAgent, and WKM.

Higher Accuracy, Lower Cost

84.33% (KnowSelf) > 79.85% (GPT-4o Expel) with far fewer knowledge queries.

Scales Across Model Sizes

Gemma-2B (2B params) with KnowSelf matches GPT-4o Expel — self-awareness is model-agnostic.

KnowSelf Efficiency

15–26% knowledge → 80–84% accuracy

Key Takeaways: Self-Awareness Is the Missing Piece

1

Indiscriminate Knowledge Is Wasteful

Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf — blanket injection hurts performance.



2

Lightweight Self-Awareness Learning

Special tokens + heuristic labeling on self-explored trajectories — no expensive human annotation or massive external feedback.



3

Outperforms Strong Baselines

KnowSelf Llama-8B: **84.33%** vs. GPT-4o ReAct **76.12%** and GPT-4o ExpeL **79.85%** — smaller model, better results.



4

Minimal Knowledge, Maximum Impact

Only **15–26%** knowledge usage vs. 100% for competing methods. Selective retrieval is both cheaper and more effective.



5

Generalizes Across Model Scales

Even **Gemma-2B** reaches **79.85%** with KnowSelf, matching GPT-4o ExpeL. The three-situation taxonomy provides a principled, model-agnostic framework for autonomous cognitive resource regulation.



Future Directions

- Extend the three-situation framework to multi-agent collaboration and embodied AI tasks
- Investigate adaptive knowledge retrieval strategies beyond binary special-token gating
- Apply metacognitive self-awareness to other domains: code generation, theorem proving, scientific reasoning

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